复变函数第零次作业

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注: 题目按照教材《复变函数简明教程》配套的全部习题的顺序排列,题号右上角有星号的题目表示出现在布置的作业题中.

1.1. 将下面的复数表示为 $a+\mathrm{i}b$ 的形式:

$$\mathrm{i}^n,\quad (1+\mathrm{i}\sqrt{3})^n,\quad (1+\mathrm{i})^n+(1-\mathrm{i})^n.$$

解答:
$\mathrm{i}^n=\left(\cos\dfrac{\pi}{2}+\mathrm{i}\sin\dfrac{\pi}{2}\right)^n=\cos\dfrac{n\pi}{2}+\mathrm{i}\sin\dfrac{n\pi}{2}$.
$(1+\mathrm{i}\sqrt{3})^n=2^n\left(\cos\dfrac{\pi}{3}+\mathrm{i}\sin\dfrac{\pi}{3}\right)^n=2^n\left(\cos\dfrac{n\pi}{3}+\mathrm{i}\sin\dfrac{n\pi}{3}\right)$.
$(1+\mathrm{i})^n+(1-\mathrm{i})^n=(\sqrt{2})^n\left(\cos\dfrac{\pi}{4}+\mathrm{i}\sin\dfrac{\pi}{4}\right)^n+(\sqrt{2})^n\left(\cos\left(-\dfrac{\pi}{4}\right)+\mathrm{i}\sin\left(-\dfrac{\pi}{4}\right)\right)^n=2(\sqrt{2})^n\cos\dfrac{n\pi}{4}$.

1.2. 解方程 $z^5=1-\mathrm{i}, z\in\mathbb{C}$.
解答: $z^5=1-\mathrm{i}=\sqrt{2}\left(\cos\left(-\dfrac{\pi}{4}\right)+\mathrm{i}\sin\left(-\dfrac{\pi}{4}\right)\right)$,
故第 k 个复根

$$z_k=\sqrt[10]{2}\left(\cos\left(-\frac{\pi}{20}+\frac{2k\pi}{5}\right)+\mathrm{i}\sin\left(-\frac{\pi}{20}+\frac{2k\pi}{5}\right)\right), 0\leqslant k\leqslant 4.$$

1.3. 设 $r,x,y\in\mathbb{R}$, $r>0$,将复数 $r^{x+\mathrm{i}y}$ 表示为 $a+\mathrm{i}b$ 的形式.
解答: $r^{x+\mathrm{i}y}=r^x\cdot r^{\mathrm{i}y}=r^x\cdot \mathrm{e}^{\mathrm{i}y\ln r}=r^x(\cos(y\ln r)+\mathrm{i}\sin(y\ln r))$.

1.4. 设 $z_1,z_2\in\mathbb{C}$, 求证: $|z_1-z_2|^2=|z_1|^2-2\mathrm{Re}z_1\bar{z}_2+|z_2|^2$,并说明其几何意义.
解答: $|z_1-z_2|^2=(z_1-z_2)(\bar{z}_1-\bar{z}_2)=z_1\bar{z}_1-z_1\bar{z}_2-z_2\bar{z}_1+z_2\bar{z}_2=|z_1|^2-2\mathrm{Re}z_1\bar{z}_2+|z_2|^2$,即为余弦定理.

1.5. 设 z_1,z_2,z_3 为单位复数,求证: z_1,z_2,z_3 为一个正三角形的顶点的充分必要条件是 $z_1+z_2+z_3=0$.
解答: 设以 z_1,z_2,z_3 为顶点的三角形(含退化情况)为 Δ ,则 Δ 的重心为 $\dfrac{1}{3}(z_1+z_2+z_3)$,垂心为 $z_1+z_2+z_3$.
故 Δ 为正三角形 $\Leftrightarrow$ Δ 的重心和垂心重合 $\Leftrightarrow$ $\dfrac{1}{3}(z_1+z_2+z_3)=z_1+z_2+z_3\Leftrightarrow z_1+z_2+z_3=0$.

1.6. 设 $z_1,z_2\in\mathbb{C}$,求证:
(1) $|1-\bar{z}_1z_2|^2-|z_1-z_2|^2=(1-|z_1|^2)(1-|z_2|^2)$;
(2)当 $|z_1|,|z_2|<1$ 时, $\left|\dfrac{z_1-z_2}{1-\bar{z}_1z_2}\right|<1$;
(3)当 $|z_1|=1$ 或 $|z_2|=1$ 且 $z_1\neq z_2$ 时, $\left|\dfrac{z_1-z_2}{1-\bar{z}_1z_2}\right|=1$.
解答:
(1)

$$\begin{aligned}|1-\bar{z}_1z_2|^2-|z_1-z_2|^2&=(1-\bar{z}_1z_2)(1-z_1\bar{z}_2)-(z_1-z_2)(\bar{z}_1-\bar{z}_2)\\&=1-\bar{z}_1z_2-z_1\bar{z}_2+z_1\bar{z}_1z_2\bar{z}_2-(z_1\bar{z}_1+z_2\bar{z}_2-z_1\bar{z}_2-\bar{z}_1z_2)\\&=1+|z_1|^2|z_2|^2-|z_1|^2-|z_2|^2=(1-|z_1|^2)(1-|z_2|^2).\end{aligned}$$

(2)由(1)知 $|1-\bar{z}_1z_2|^2>|z_1-z_2|^2$,故只需证: $|1-\bar{z}_1z_2|\neq 0$, 即$\bar{z}_1z_2\neq 1$.
事实上, $|\bar{z}_1z_2|=|z_1||z_2|<1\cdot 1=1$,故 $\bar{z}_1z_2\neq 1$.
(3)不妨设 $|z_1|=1$,由(1)知 $|1-\bar{z}_1z_2|^2=|z_1-z_2|^2$,故只需证: $|1-\bar{z}_1z_2|\neq 0$,即 $1-\bar{z}_1z_2\neq 0$.
事实上, $1-\bar{z}_1z_2=|z_1|^2-\bar{z}_1z_2=z_1\bar{z}_1-\bar{z}_1z_2=\bar{z}_1(z_1-z_2)\neq 0$.

1.7. 在复平面上,求过 $1+3\mathrm{i}$ 和 $-1+4\mathrm{i}$ 的直线的方程.
解答: 记 $z_1=1+3\mathrm{i}, z_2=-1+4\mathrm{i}$,则过 z_1 和 z_2 的直线的方程为 $(z-z_1)(\bar{z}-\bar{z}_2)=(\bar{z}-\bar{z}_1)(z-z_2)$.

1.8.
(1)设 $A,C\in\mathbb{R},B\in\mathbb{C}$,问方程 $Az\bar{z}+B\bar{z}+\bar{B}z+C=0$ 在什么条件下是圆方程,并求其圆心和半径;
(2)在上面的方程中令 $A\to 0$,求半径和圆心的极限,并说明其几何意义.
解答:
(1)显然 $A\neq 0$,则

$$Az\bar{z}+B\bar{z}+\bar{B}z+C=0\Leftrightarrow\left(z+\frac{B}{A}\right)\left(\bar{z}+\frac{\bar{B}}{A}\right)=\frac{|B|^2-AC}{A^2}\Leftrightarrow\left|z+\frac{B}{A}\right|=\frac{\sqrt{|B|^2-AC}}{|A|}.$$

故 $|B|^2>AC$,且圆心为 $-\dfrac{B}{A}$,半径为 $\dfrac{\sqrt{|B|^2-AC}}{|A|}$.
(2)当 $B\neq 0$ 时,圆心 $\to\infty$,半径 $\to+\infty$;当 $B=0$ 时,圆心 $=0$,半径 $\to+\infty$.即为圆退化成直线.

1.9. 设 $C\in\mathbb{R},B\in\mathbb{C}$,求证: 直线 $B\bar{z}+\bar{B}z+C=0$ 是 z_1,z_2 连线的垂直平分线的充分必要条件是

$$B\bar{z}_1+\bar{B}z_2+C=0.$$

解答: z_1,z_2 连线的垂直平分线的方程为

$$|z-z_1|=|z-z_2|\Leftrightarrow(z-z_1)(\bar{z}-\bar{z}_1)=(z-z_2)(\bar{z}-\bar{z}_2)\Leftrightarrow(\bar{z}_1-\bar{z}_2)z+(z_1-z_2)\bar{z}+(z_2\bar{z}_2-z_1\bar{z}_1)=0.$$

故直线 $B\bar{z}+\bar{B}z+C=0$ 是 z_1,z_2 连线的垂直平分线 $\Leftrightarrow C=\dfrac{z_2\bar{z}_2-z_1\bar{z}_1}{z_1-z_2}B$.
并且

$$B\bar{z}_1+\bar{B}z_2+C=0\Leftrightarrow\begin{cases}B\bar{z}_1+\bar{B}z_2+C=0\\ \bar{B}z_1+B\bar{z}_2+C=0\end{cases}\Leftrightarrow\begin{cases}C=\dfrac{z_2\bar{z}_2-z_1\bar{z}_1}{z_1-z_2}B\\ C=\dfrac{z_2\bar{z}_2-z_1\bar{z}_1}{\bar{z}_1-\bar{z}_2}\bar{B}\end{cases}\Leftrightarrow C=\frac{z_2\bar{z}_2-z_1\bar{z}_1}{z_1-z_2}B.$$

1.10. 设 $S\subseteq\mathbb{R}^n$,求证:
(1) S' 为闭集;
(2) $S\cup S'=\bigcap\limits_{\text{闭集}T\supseteq S}T\quad(=\overline{S})$.
解答:
(1)由《数学分析(第三册)》习题**13.8**的(1)即证.
(2) $\forall$闭集$T\supseteq S$, $\forall\boldsymbol{x}\in S'$, $\exists$ 互异的点列 $\{\boldsymbol{x}_k\}\subseteq S$,使得 $\boldsymbol{x}_k\to\boldsymbol{x}$,又 $\boldsymbol{x}_k\in S\subseteq T$,且 T 为闭集,故 $\boldsymbol{x}\in T$.
从而 $S'\subseteq T$.因此 $S\cup S'\subseteq T$,故 $S\cup S'\subseteq\bigcap\limits_{\text{闭集}T\supseteq S}T$.
再由《数学分析(第三册)》习题**13.6**的(1)知 $S\cup S'=((S^c)^\circ)^c$ 为闭集,且 $(S\cup S')\supseteq S$,故 $\bigcap\limits_{\text{闭集}T\supseteq S}T\subseteq S\cup S'$.
因此 $S\cup S'=\bigcap\limits_{\text{闭集}T\supseteq S}T$.

1.11. 设 $S\subseteq\mathbb{R}^n$,求证: S 为紧集 $\Leftrightarrow S$ 为有界闭集.

解答: 熟知.

1.12. 设 $S \subseteq \mathbb{R}^n$,求证: $\mathrm{diam}S = \mathrm{diam}\overline{S}$.
解答: 由于 $S \subseteq \overline{S}$,故 $\mathrm{diam}S \leqslant \mathrm{diam}\overline{S}$.
若 $\mathrm{diam}\overline{S} = 0$,则 $\mathrm{diam}S = 0 = \mathrm{diam}\overline{S}$.下设 $\mathrm{diam}\overline{S} = d > 0$,则 $\mathrm{diam}S \leqslant d$.
$\forall \varepsilon > 0$,由于 $\mathrm{diam}\overline{S} = \sup\{|\boldsymbol{x} - \boldsymbol{y}| : \boldsymbol{x}, \boldsymbol{y} \in \overline{S}\} = d$,故 $\exists \boldsymbol{x}_1, \boldsymbol{y}_1 \in \overline{S}$,使得 $|\boldsymbol{x}_1 - \boldsymbol{y}_1| \geqslant d - \dfrac{\varepsilon}{3}$.
又由于 $\boldsymbol{x}_1, \boldsymbol{y}_1 \in \overline{S}$,故 $U\left(\boldsymbol{x}_1, \dfrac{\varepsilon}{3}\right) \cap S \neq \varnothing$, $U\left(\boldsymbol{y}_1, \dfrac{\varepsilon}{3}\right) \cap S \neq \varnothing$,即 $\exists \boldsymbol{x}_2, \boldsymbol{y}_2 \in S$,使得 $|\boldsymbol{x}_1 - \boldsymbol{x}_2|, |\boldsymbol{y}_1 - \boldsymbol{y}_2| < \dfrac{\varepsilon}{3}$.
从而

$$|\boldsymbol{x}_2 - \boldsymbol{y}_2| \geqslant |\boldsymbol{x}_1 - \boldsymbol{y}_1| - |\boldsymbol{x}_1 - \boldsymbol{x}_2| - |\boldsymbol{y}_1 - \boldsymbol{y}_2| > \left(d - \frac{\varepsilon}{3}\right) - \frac{\varepsilon}{3} - \frac{\varepsilon}{3} = d - \varepsilon.$$

因此 $\mathrm{diam}S = \sup\{|\boldsymbol{x} - \boldsymbol{y}| : \boldsymbol{x}, \boldsymbol{y} \in S\} \geqslant d$.故 $\mathrm{diam}S = d = \mathrm{diam}\overline{S}$.

1.13. 设 $z_0 \in \mathbb{C}$, $\lim\limits_{n\to+\infty} z_n = z_0$.
(1)求证: $\lim\limits_{n\to+\infty} |z_n| = |z_0|$;
(2)若 $z_0 \neq 0$,求证: $\lim\limits_{n\to+\infty} \mathrm{Arg}(z_n) = \mathrm{Arg}(z_0)$.
解答:
(1) $\forall \varepsilon > 0$, $\exists N \in \mathbb{Z}^+$,使得 $\forall n \geqslant N$,都有 $|z_n - z_0| < \varepsilon$,从而 $||z_n| - |z_0|| \leqslant |z_n - z_0| < \varepsilon$,因此 $\lim\limits_{n\to+\infty} |z_n| = |z_0|$.
(2)由旋转不变性和齐次性,不妨设 $z_0 = 1$,则 $\mathrm{Arg}(z_0) = 0$.
$\forall n \in \mathbb{Z}^+$,设 $z_n = r_n \mathrm{e}^{\mathrm{i}\theta_n}$,其中 $r_n = |z_n| \geqslant 0$, $\theta_n = \mathrm{Arg}(z_n) \in (-\pi, \pi]$,则

$$|z_n - z_0| = |r_n \mathrm{e}^{\mathrm{i}\theta_n} - 1| = (r_n \mathrm{e}^{\mathrm{i}\theta_n} - 1)(r_n \mathrm{e}^{-\mathrm{i}\theta_n} - 1) = r_n^2 - 2r_n \cos\theta_n + 1.$$

又由(1)知 $\lim\limits_{n\to+\infty} r_n = 1$,故

$$\lim_{n\to+\infty} |\theta_n| = \lim_{n\to+\infty} \arccos(\cos\theta_n) = \lim_{n\to+\infty} \arccos\left(\frac{r_n^2 + 1 - |z_n - z_0|}{2r_n}\right) = \arccos\left(\frac{1 + 1 - 0}{2}\right) = 0.$$

因此 $\lim\limits_{n\to+\infty} \mathrm{Arg}(z_n) = \lim\limits_{n\to+\infty} \theta_n = 0 = \mathrm{Arg}(z_0)$.

1.14. 设 $S \subseteq \mathbb{R}^n$ 为开集,求证: S 可表示为至多可数个互不相交且连通的开集的并.
解答:

1.15. 设 $S \subseteq \mathbb{R}^n$. $T \subseteq S$ 称为 S 的 **相对闭集** ,如果 $(T' \cap S) \subseteq T$; $T \subseteq S$ 称为 S 的 **相对开集** ,如果 $S \backslash T$ 是 S 的相对闭集.求证: S 连通的充要条件是 S 不能分解为两个非空且不交的相对开集(闭集)的并.
解答:

1.16. 设 $S \subseteq \mathbb{R}^n$ 连通, $f(\boldsymbol{x})$ 定义在 S 上.如果 $\forall \boldsymbol{x}_0 \in S$, $\exists r > 0$,使得 $f(\boldsymbol{x})$ 在 $S \cap U(\boldsymbol{x}_0, r)$ 上为常数,求证: $f(\boldsymbol{x})$ 在 S 上为常数.
解答:

1.17. 设 $U, V \subseteq \mathbb{R}^n$ 是区域.映射 $f : U \to V$ 称为 **开映射** ,如果 $\forall$ 开集 $U_1 \subseteq U$,都有 $f(U_1) \subseteq V$ 为开集; f 称为 **逆紧** 的,如果 $\forall$ 紧集 $K \subseteq V$,都有 $f^{-1}(K) \subseteq U$ 为紧集.求证: 如果 f 是开且逆紧的映射,则 $f(U) = V$.
解答:

1.18. 求 $\sum\limits_{k=0}^{n} (\cos k\theta + \mathrm{i}\sin k\theta)$.
解答: $\sum\limits_{k=0}^{n} (\cos k\theta + \mathrm{i}\sin k\theta) = \sum\limits_{k=0}^{n} \mathrm{e}^{\mathrm{i}k\theta} = \begin{cases} n+1, & \theta = 2m\pi, m \in \mathbb{Z} \\ \dfrac{1 - \mathrm{e}^{\mathrm{i}(n+1)\theta}}{1 - \mathrm{e}^{\mathrm{i}\theta}}, & \theta \neq 2m\pi, m \in \mathbb{Z} \end{cases}$.

1.19. 求证:
(1)若 $K \subseteq \mathbb{R}^n$ 为紧集, $F \subseteq \mathbb{R}^n$ 为闭集,且 $K \cap F = \varnothing$,则 $\mathrm{d}(K, F) > 0$;
(2)若 $D \subseteq \mathbb{R}^n$ 为开集, $S \subseteq D$ 为紧集,则 $\mathrm{d}(S, \partial D) > 0$.
解答:

1.20. 设 $f(x, y) = x^3 + 3xy + y$,求 $\dfrac{\partial f}{\partial z}, \dfrac{\partial f}{\partial \bar{z}}$.
解答: 将 $x = \dfrac{1}{2}(z + \bar{z}), y = \dfrac{1}{2\mathrm{i}}(z - \bar{z})$ 代入 $f(x, y) = x^3 + 3xy + y$ 计算即可.

1.21. 设 $S \subseteq \mathbb{R}^n, \boldsymbol{x}_0 \in S'$,给出并证明: $S \ni \boldsymbol{x} \to \boldsymbol{x}_0$ 时,函数 $f(\boldsymbol{x})$ 收敛的 Cauchy 准则.
解答: $\forall \varepsilon > 0$, $\exists \delta > 0$,使得 $\forall \boldsymbol{x}_1, \boldsymbol{x}_2 \in S \cap U_0(\boldsymbol{x}_0, \delta)$,都有 $|f(\boldsymbol{x}_1) - f(\boldsymbol{x}_2)| < \varepsilon$.易证.

1.22. 将向量函数 $(x, y) \mapsto (u(x, y), v(x, y))$ 表示为复函数 $w = u + \mathrm{i}v = f(z) = f(x + \mathrm{i}y)$.如果 $f(x + \mathrm{i}y)$ 连续可导,求证: 映射的 Jacobi 行列式满足:

$$\begin{vmatrix} \dfrac{\partial u}{\partial x} & \dfrac{\partial u}{\partial y} \\ \dfrac{\partial v}{\partial x} & \dfrac{\partial v}{\partial y} \end{vmatrix} = \begin{vmatrix} \dfrac{\partial f}{\partial z} & \dfrac{\partial \bar{f}}{\partial z} \\ \dfrac{\partial f}{\partial \bar{z}} & \dfrac{\partial \bar{f}}{\partial \bar{z}} \end{vmatrix}.$$

解答:

$$\begin{aligned}
\begin{vmatrix} \dfrac{\partial f}{\partial z} & \dfrac{\partial \bar{f}}{\partial z} \\ \dfrac{\partial f}{\partial \bar{z}} & \dfrac{\partial \bar{f}}{\partial \bar{z}} \end{vmatrix} &= \frac{\partial f}{\partial z} \cdot \frac{\partial \bar{f}}{\partial \bar{z}} - \frac{\partial \bar{f}}{\partial z} \cdot \frac{\partial f}{\partial \bar{z}} \\
&= \frac{1}{2}\left(\frac{\partial f}{\partial x} - \mathrm{i}\frac{\partial f}{\partial y}\right) \cdot \frac{1}{2}\left(\frac{\partial \bar{f}}{\partial x} + \mathrm{i}\frac{\partial \bar{f}}{\partial y}\right) - \frac{1}{2}\left(\frac{\partial \bar{f}}{\partial x} - \mathrm{i}\frac{\partial \bar{f}}{\partial y}\right) \cdot \frac{1}{2}\left(\frac{\partial f}{\partial x} + \mathrm{i}\frac{\partial f}{\partial y}\right) \\
&= \frac{1}{2}\mathrm{i}\left(\frac{\partial f}{\partial x} \cdot \frac{\partial \bar{f}}{\partial y} - \frac{\partial f}{\partial y} \cdot \frac{\partial \bar{f}}{\partial x}\right) \\
&= \frac{1}{2}\mathrm{i}\left(\left(\frac{\partial u}{\partial x} + \mathrm{i}\frac{\partial v}{\partial x}\right)\left(\frac{\partial u}{\partial y} - \mathrm{i}\frac{\partial v}{\partial y}\right) - \left(\frac{\partial u}{\partial y} + \mathrm{i}\frac{\partial v}{\partial y}\right)\left(\frac{\partial u}{\partial x} - \mathrm{i}\frac{\partial v}{\partial x}\right)\right) \\
&= \frac{1}{2}\mathrm{i} \cdot 2\mathrm{i}\left(\frac{\partial v}{\partial x} \cdot \frac{\partial u}{\partial y} - \frac{\partial u}{\partial x} \cdot \frac{\partial v}{\partial y}\right) \\
&= \frac{\partial u}{\partial x} \cdot \frac{\partial v}{\partial y} - \frac{\partial v}{\partial x} \cdot \frac{\partial u}{\partial y} = \begin{vmatrix} \dfrac{\partial u}{\partial x} & \dfrac{\partial u}{\partial y} \\ \dfrac{\partial v}{\partial x} & \dfrac{\partial v}{\partial y} \end{vmatrix}.
\end{aligned}$$

1.23. 试用 $\dfrac{\partial}{\partial z}, \dfrac{\partial}{\partial \bar{z}}$ 表示 $\dfrac{\partial^2}{\partial x^2} + \dfrac{\partial^2}{\partial x \partial y}$.
解答: 利用 $\dfrac{\partial}{\partial x} = \dfrac{\partial}{\partial z} + \dfrac{\partial}{\partial \bar{z}}, \dfrac{\partial}{\partial y} = \mathrm{i}\left(\dfrac{\partial}{\partial z} - \dfrac{\partial}{\partial \bar{z}}\right)$ 直接计算即可.